FSWD LAB 4

MARYAM JAHANGIR 241949

SUBMITTED TO:

FIZZA ASMAT

Class Practice

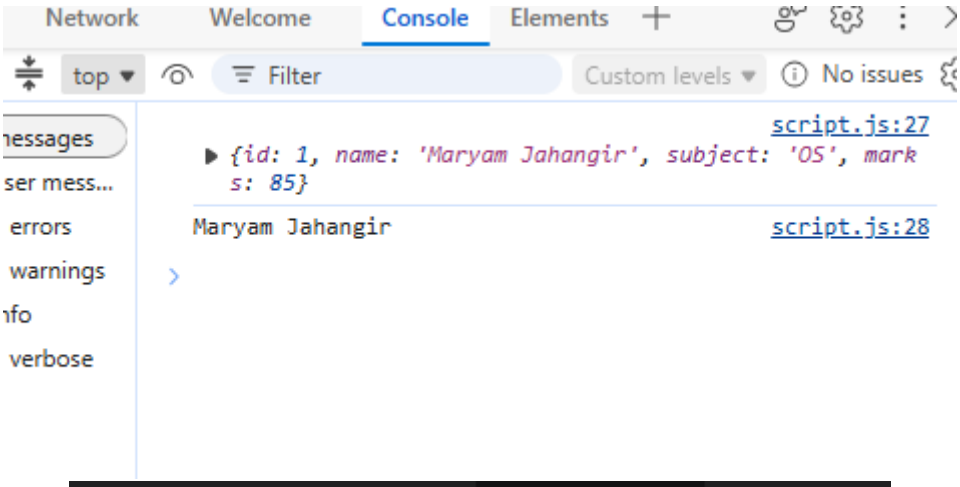
Lab4 > JS script.js > ...

```
1   document.title= "Student Result Dashboard";
2   //comments
3   /*Multi line comments */
4   console.log(name);
5   console.log("marks");
6   let name = "John Doe";
7   var marks = 85;
```

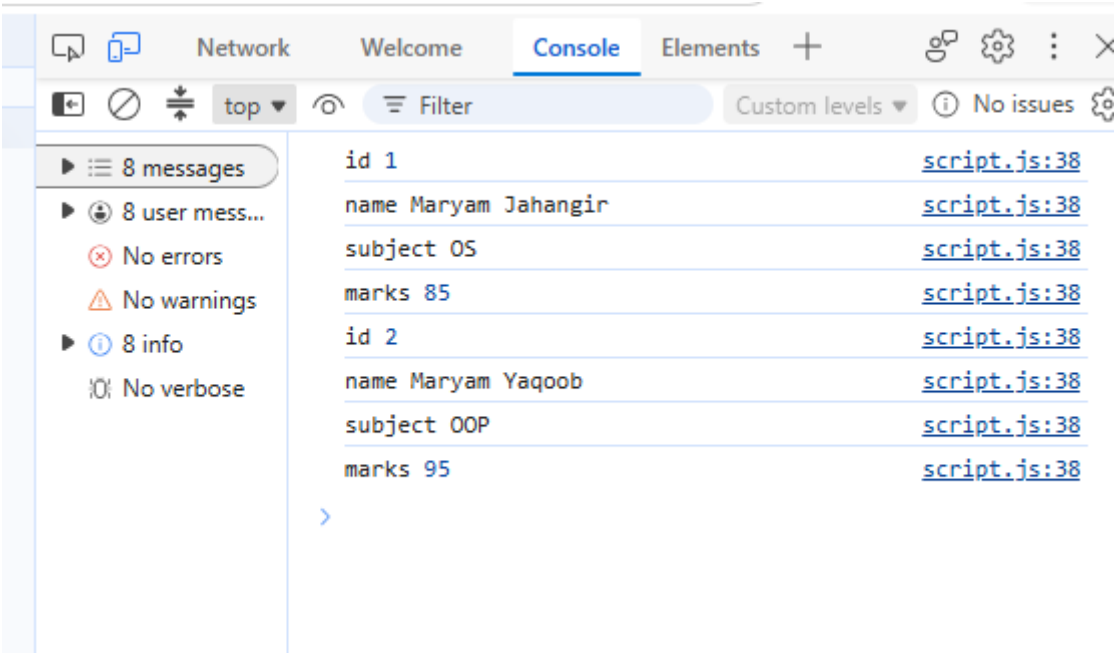
4 > JS script.js > ...

```
document.title= "Student Result Dashboard";
//comments
/*Multi line comments */
/*console.log(name);
console.log("marks");
let name = "John Doe";
var marks = 85;*/
const student ={
    id: 1,
    name: "Maryam Jahangir",
    subject: "OS",
    marks: 85
}
const arry = [{
    id: 1,
    name: "Maryam Jahangir",
    subject: "OS",
    marks: 85
},
{
    id: 2,
    name: "Maryam Yaqoob",
    subject: "OOP",
    marks: 95
}]

console.log(student);
console.log(arry[0].name);
```

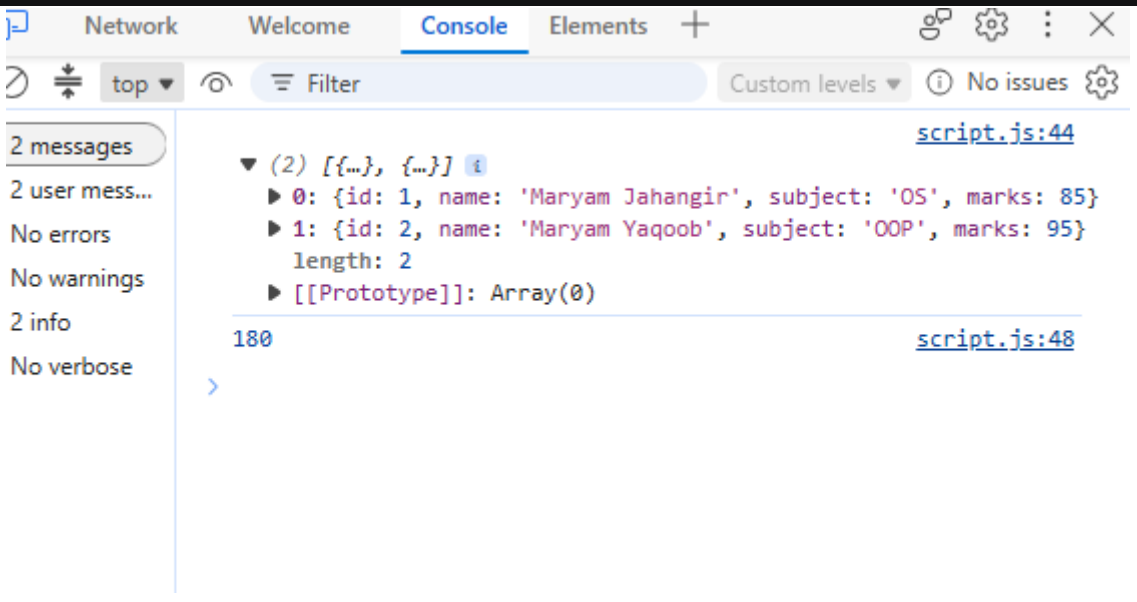



```
ab4 > JS script.js > ...
14  }
15  const arry = [{
16      id: 1,
17      name: "Maryam Jahangir",
18      subject: "OS",
19      marks: 85
20  },
21  {
22      id: 2,
23      name: "Maryam Yaqoob",
24      subject: "OOP",
25      marks: 95
26  }]
27
28  /*csole.log(student);
29  console.log(arry[0].name);
30
31  // for---in  loop in a single objects
32  /*for( const key in student){
33      console.log(key, student[key]);
34  }   */
35
36  for(const i of arry){
37      for(const key in i){
38          console.log(key, i[key]);
39      }
40  }
41
```




```
    }
}*/
//filter method
const passing = arry.filter(i => i.marks >= 40);
console.log(passing);

//reduce  perform calculation on array of objects and return single value
const total =arry.reduce((sum, i) => sum + i.marks, 0);
console.log(total);
```

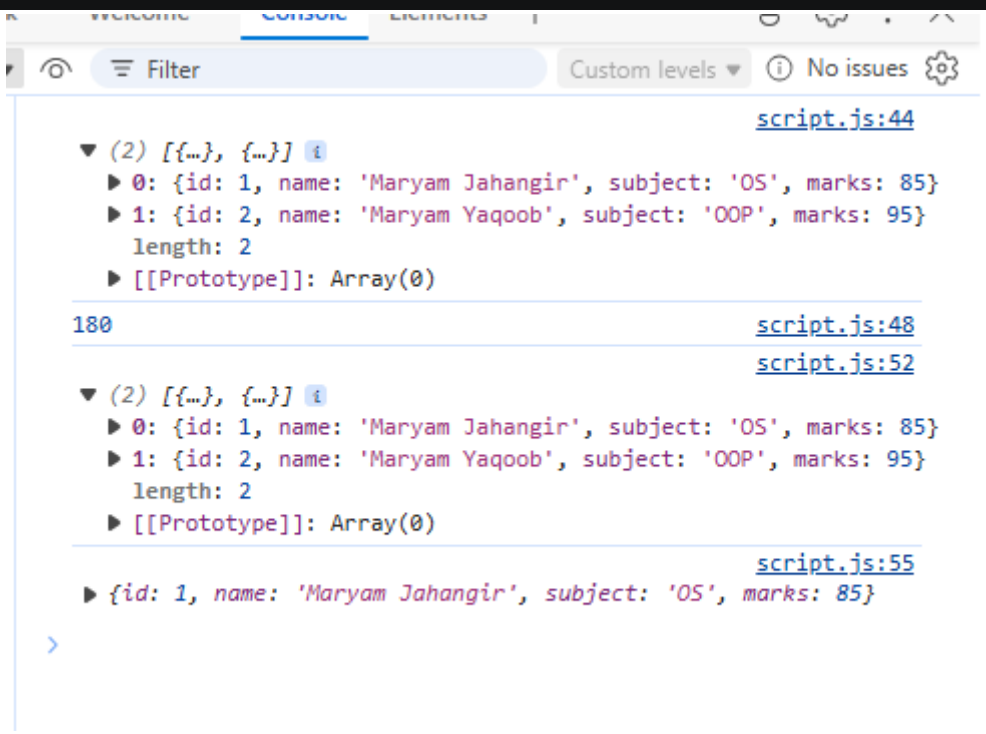


```
}   */
    /*or---of loop in an array of objects
for(const i of arry){
    for(const key in i){
        console.log(key, i[key]);
    }
}*/
//filter method
const passing = arry.filter(i => i.marks >= 40);
console.log(passing);

//reduce  perform calculation on array of objects and return single val
const total =arry.reduce((sum, i) => sum + i.marks, 0);
console.log(total);

// sort method
const sorted = arry.sort((a, b) => a.marks - b.marks);
console.log(sorted);
 // find method

 const find = arry.find(i => i.name === "Maryam Jahangir");
 console.log(find);
```



index.html # style.css JS script.js M

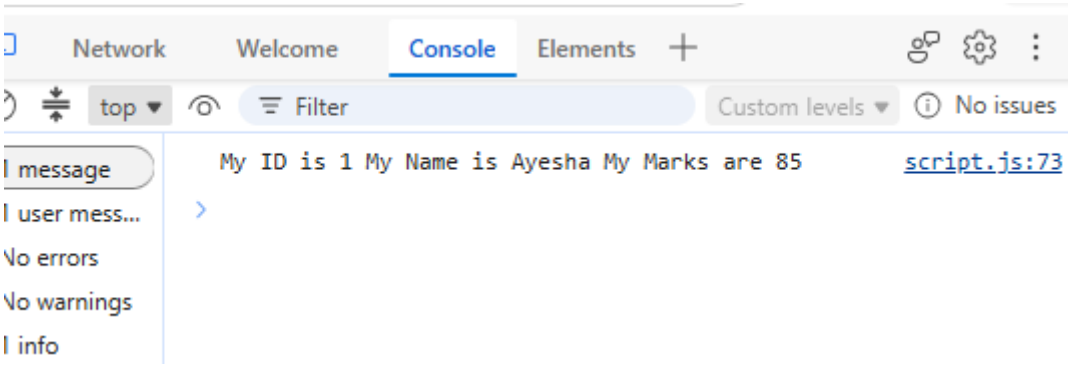
Lab4 > JS script.js > ...

```
//reduce  perform calculation on array of objects and return single value
const total =arry.reduce((sum, i) => sum + i.marks, 0);
console.log(total);

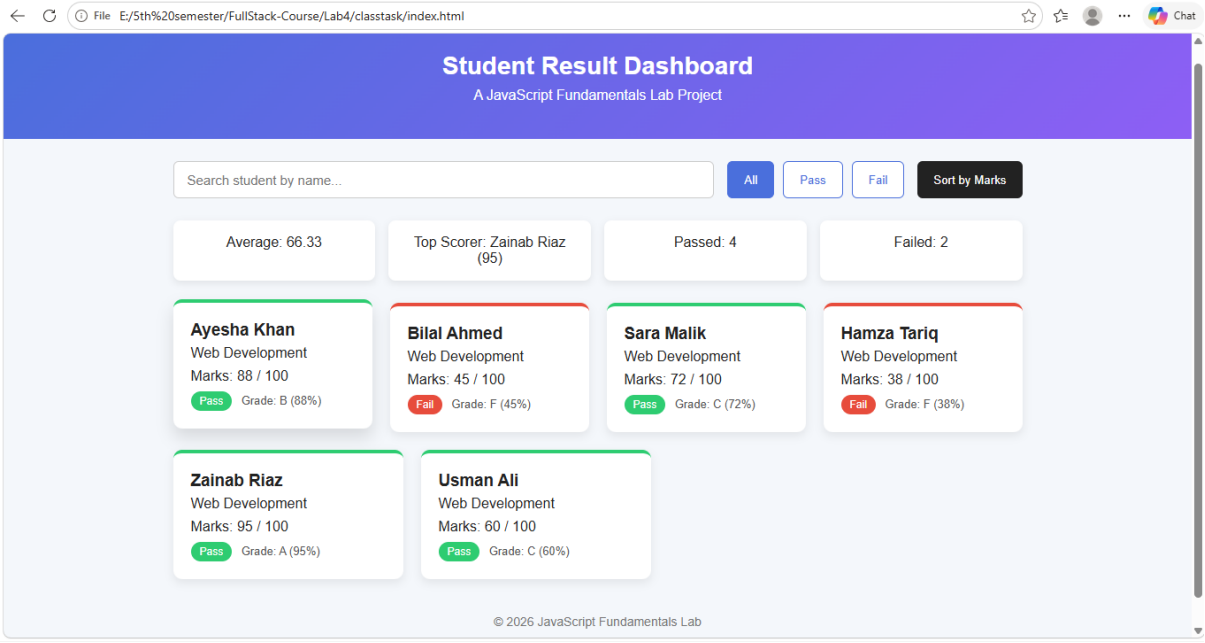
// sort method
const sorted = arry.sort((a, b) => a.marks - b.marks);
console.log(sorted);
 // find method

 const find = arry.find(i => i.name === "Maryam Jahangir");
 console.log(find);

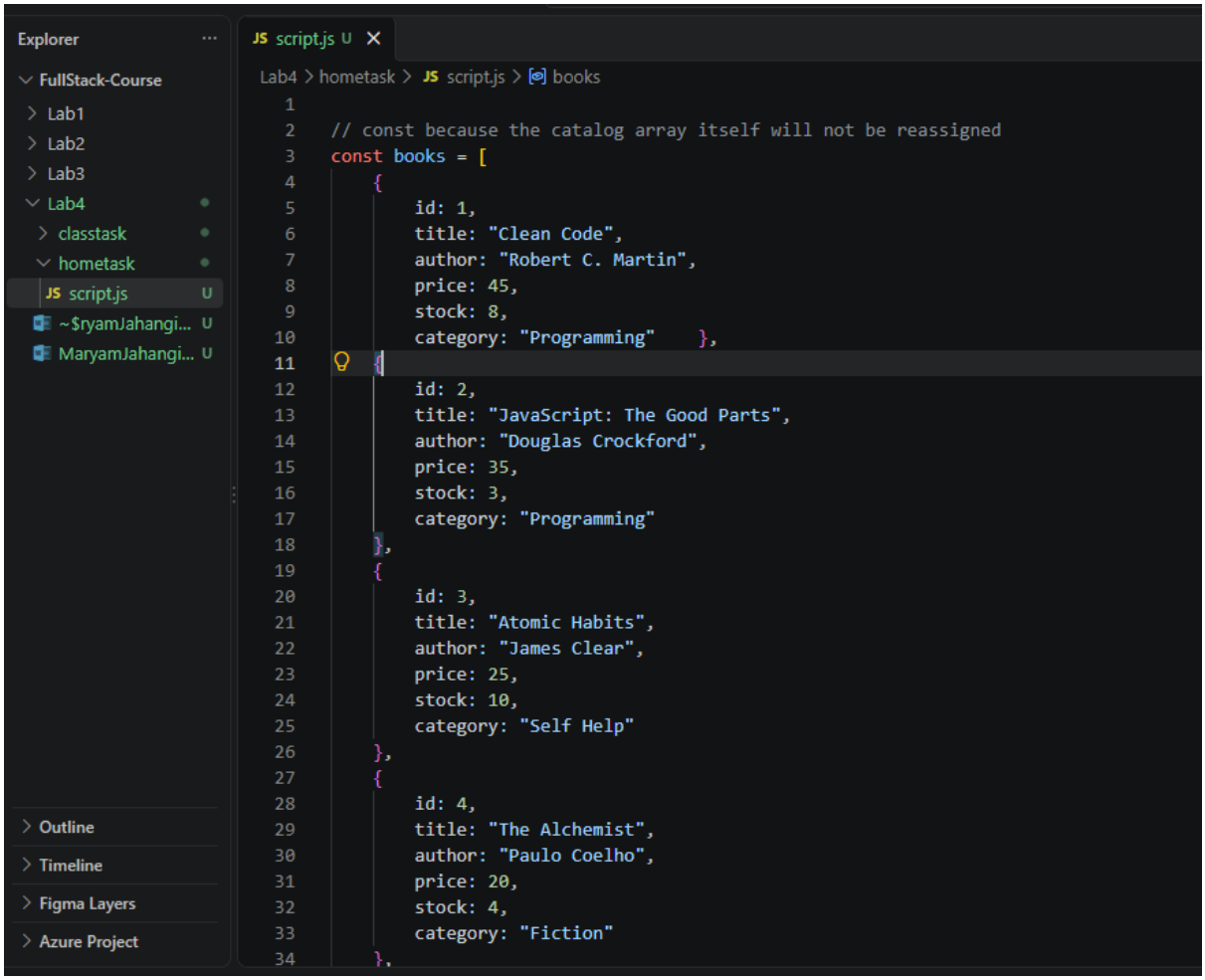
 */
class Student{
    constructor(id, name, marks){
        this.id= id;
        this.name= name;
        this.marks= marks;
    }
    getstatus(){
        return this.marks >= 40 ? "Pass" : "Fail";
    }
}
const std=new Student(1, "Ayesha", 85);
std.getstatus();
const {id, name, marks} = std;
console.log("My ID is "+id+" My Name is "+name+" My Marks are "+marks);
```

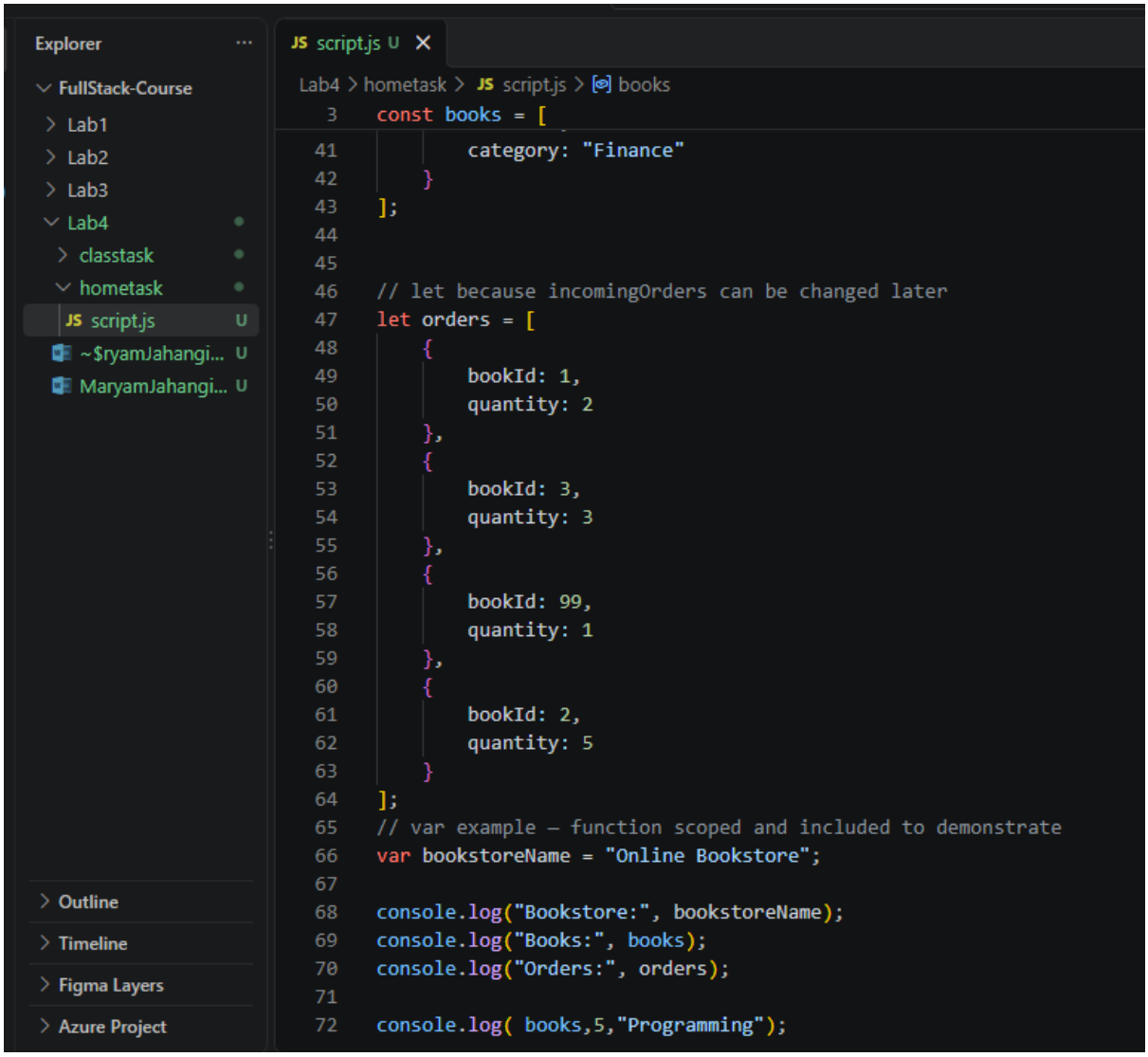


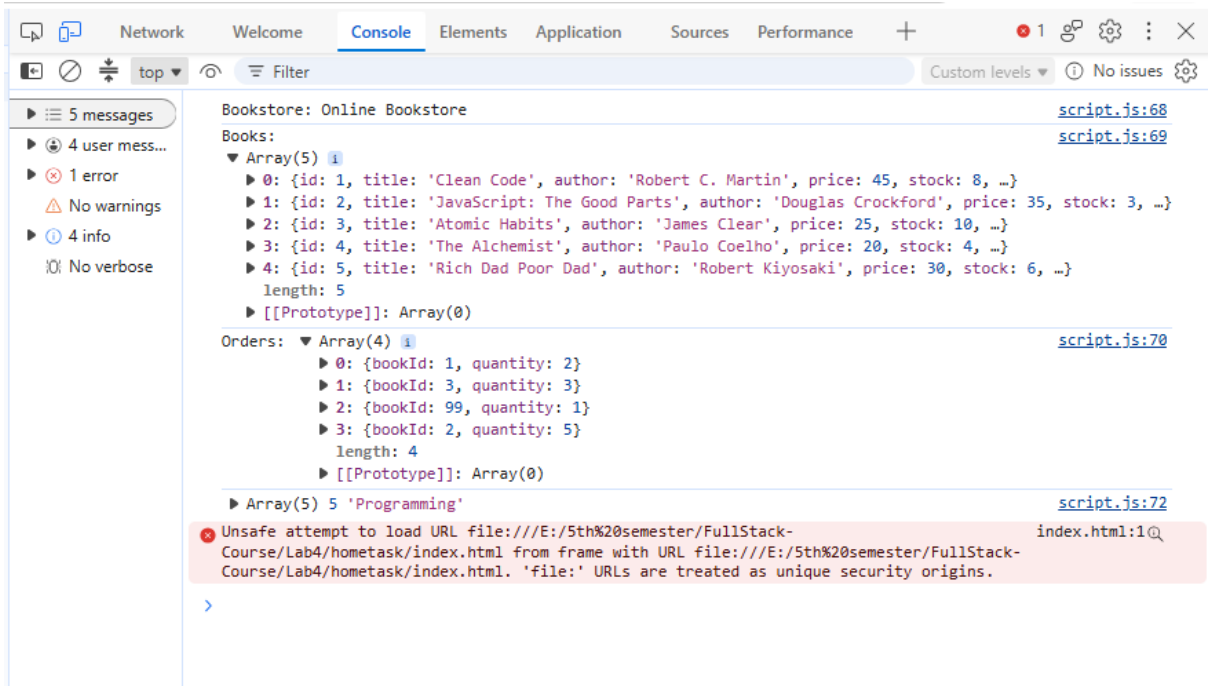
```
> JS script.js > ...
class Student {
    // CONDITIONS: if / else if / else
    getGrade() {
        if (this.marks >= 90) {
            return "A";
        } else if (this.marks >= 75) {
            return "B";
        } else if (this.marks >= 50) {
            return "C";
        } else {
            return "F";
        }
    }
}
```



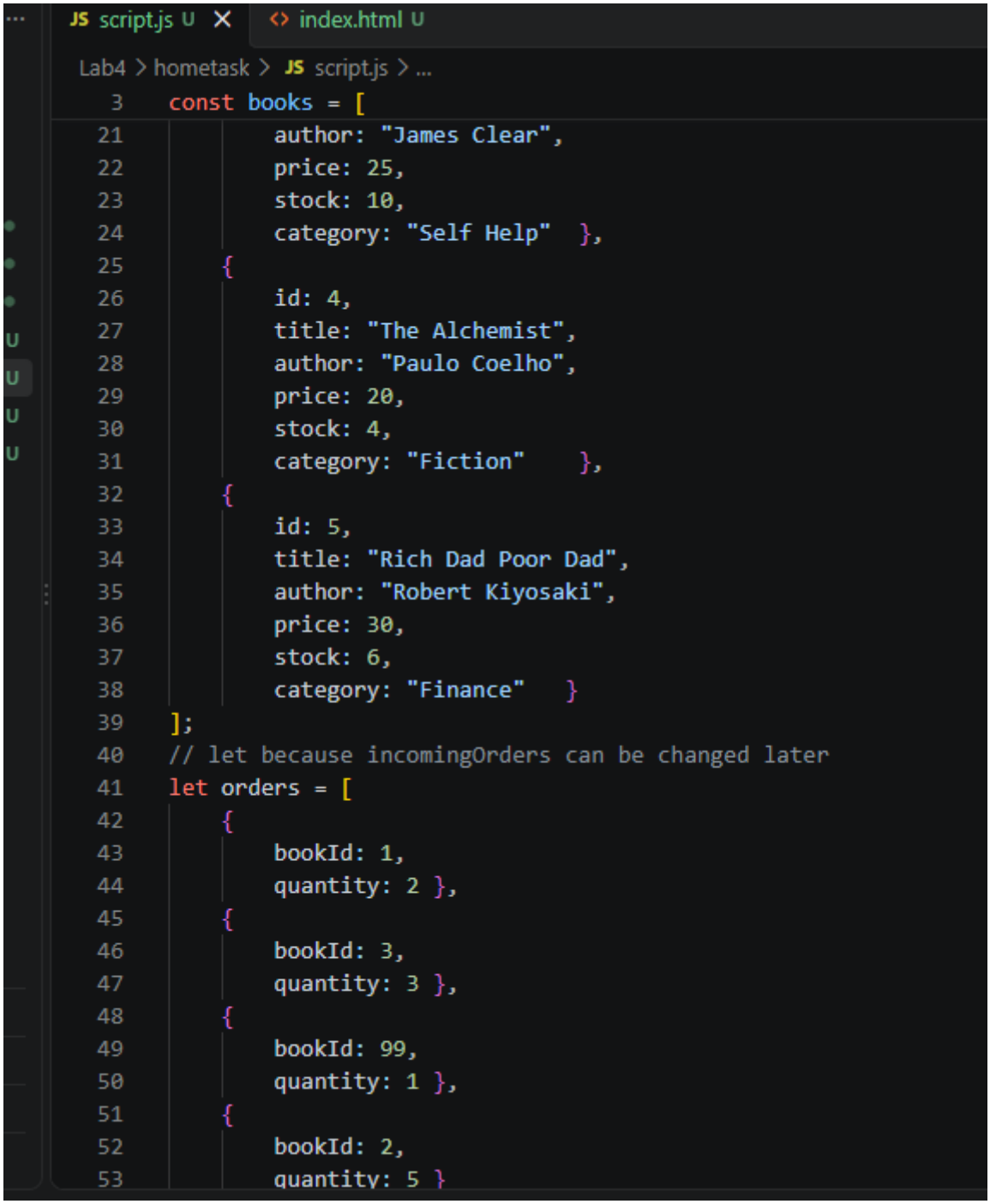
Task 1


Explorer
FullStack-Course
Lab1
Lab2
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Lab4
classtask
hometask
script.js
~$ryamJahangi...
MaryamJahangi...
Outline
Timeline
Figma Layers
Azure Project
script.js U
Lab4 > hometask > script.js > books
// const because the catalog array itself will not be reassigned
const books = [
{
id: 1,
title: "Clean Code",
author: "Robert C. Martin",
price: 45,
stock: 8,
category: "Programming" },
{
id: 2,
title: "JavaScript: The Good Parts",
author: "Douglas Crockford",
price: 35,
stock: 3,
category: "Programming"
},
{
id: 3,
title: "Atomic Habits",
author: "James Clear",
price: 25,
stock: 10,
category: "Self Help"
},
{
id: 4,
title: "The Alchemist",
author: "Paulo Coelho",
price: 20,
stock: 4,
category: "Fiction"
},


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~$ryamJahangi... U
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Outline
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Figma Layers
Azure Project
Lab4 > hometask > script.js > books
3 const books = [
41 category: "Finance"
42 }
43];
44
45
46 // let because incomingOrders can be changed later
47 let orders = [
48 {
49 bookId: 1,
50 quantity: 2
51 },
52 {
53 bookId: 3,
54 quantity: 3
55 },
56 {
57 bookId: 99,
58 quantity: 1
59 },
60 {
61 bookId: 2,
62 quantity: 5
63 }
64];
65 // var example – function scoped and included to demonstrate
66 var bookstoreName = "Online Bookstore";
67
68 console.log("Bookstore:", bookstoreName);
69 console.log("Books:", books);
70 console.log("Orders:", orders);
71
72 console.log(books,5,"Programming");


Network
Welcome
Console
Elements
Application
Sources
Performance
top
Filter
Custom levels
No issues
5 messages
4 user mess...
1 error
No warnings
4 info
No verbose
Bookstore: Online Bookstore
script.js:68
Books:
script.js:69
Array(5)
0: {id: 1, title: 'Clean Code', author: 'Robert C. Martin', price: 45, stock: 8, …}
1: {id: 2, title: 'JavaScript: The Good Parts', author: 'Douglas Crockford', price: 35, stock: 3, …}
2: {id: 3, title: 'Atomic Habits', author: 'James Clear', price: 25, stock: 10, …}
3: {id: 4, title: 'The Alchemist', author: 'Paulo Coelho', price: 20, stock: 4, …}
4: {id: 5, title: 'Rich Dad Poor Dad', author: 'Robert Kiyosaki', price: 30, stock: 6, …}
length: 5
[[Prototype]]: Array(0)
Orders:
script.js:70
Array(4)
0: {bookId: 1, quantity: 2}
1: {bookId: 3, quantity: 3}
2: {bookId: 99, quantity: 1}
3: {bookId: 2, quantity: 5}
length: 4
[[Prototype]]: Array(0)
Array(5) 5 'Programming'
script.js:72
Unsafe attempt to load URL file:///E:/5th%20semester/FullStack-Course/Lab4/hometask/index.html from frame with URL file:///E:/5th%20semester/FullStack-Course/Lab4/hometask/index.html. 'file:' URLs are treated as unique security origins.
index.html:1

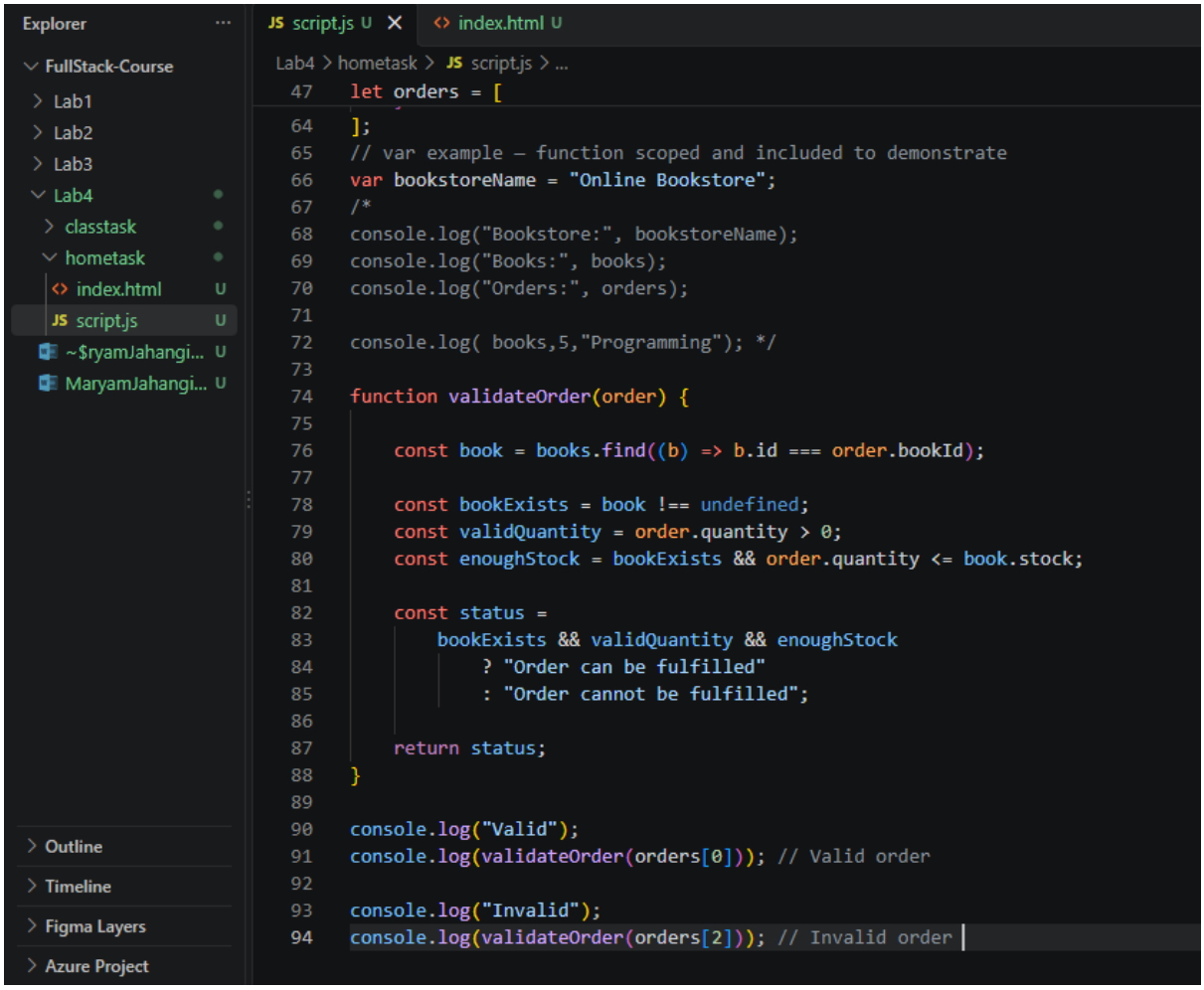
Task 2

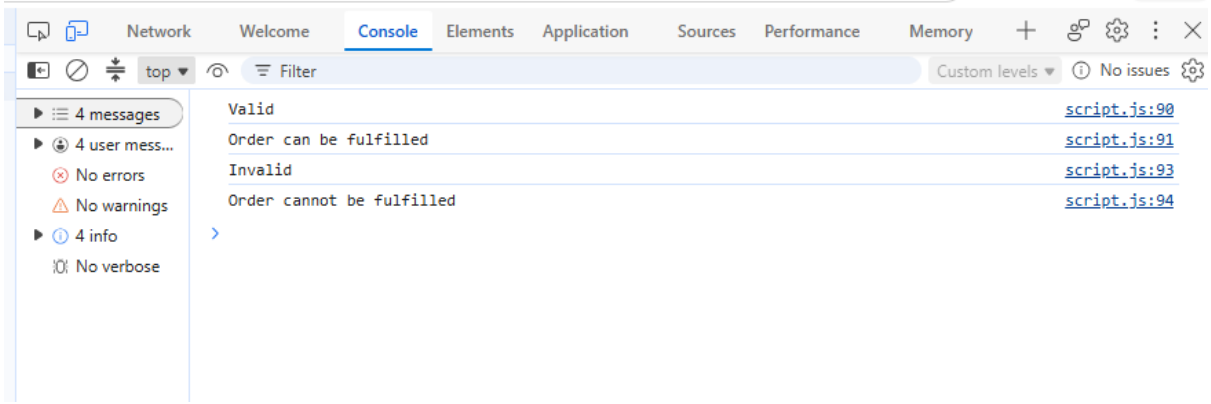


JS script.js U <> index.html U

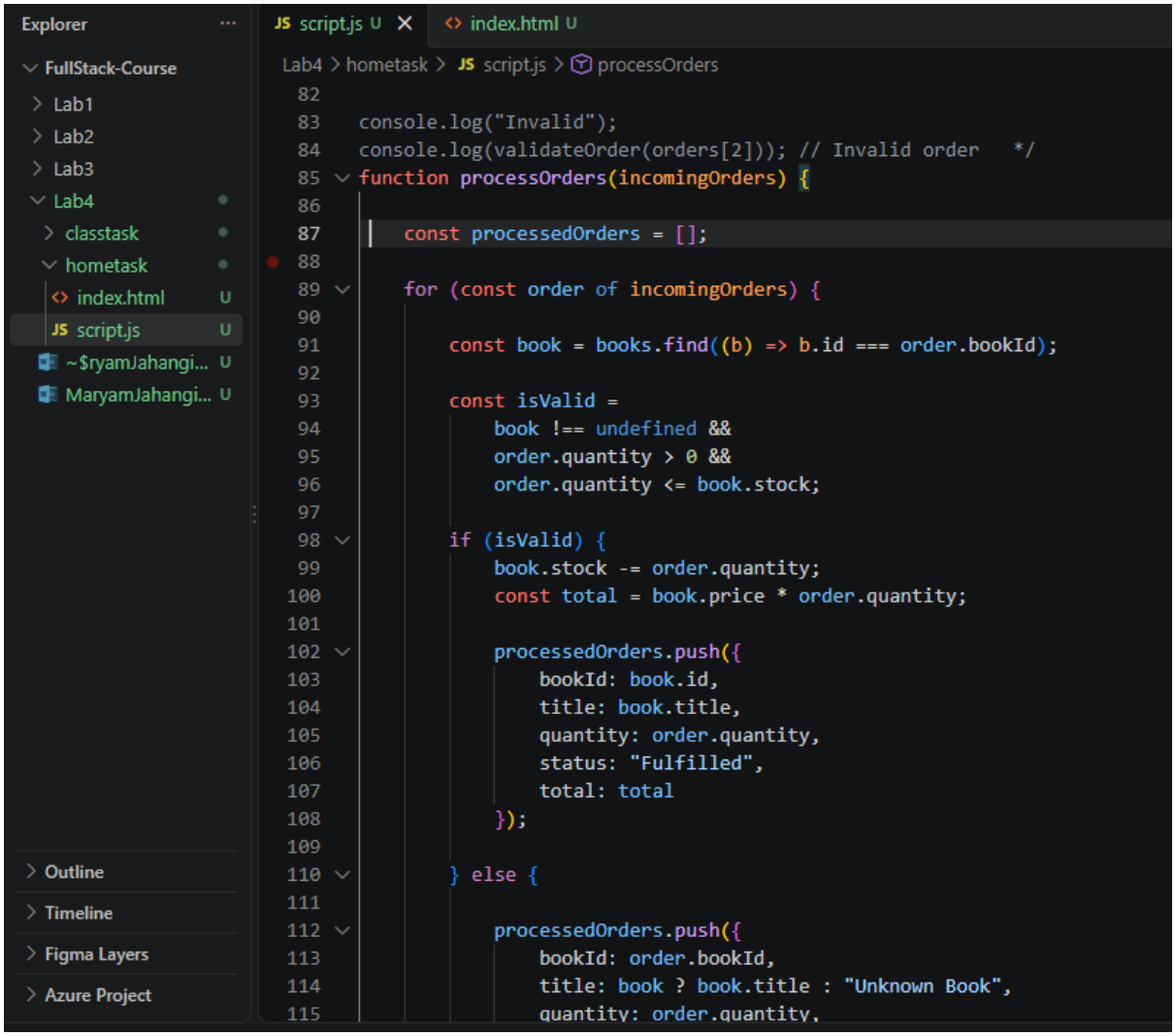
Lab4 > hometask > JS script.js > ...

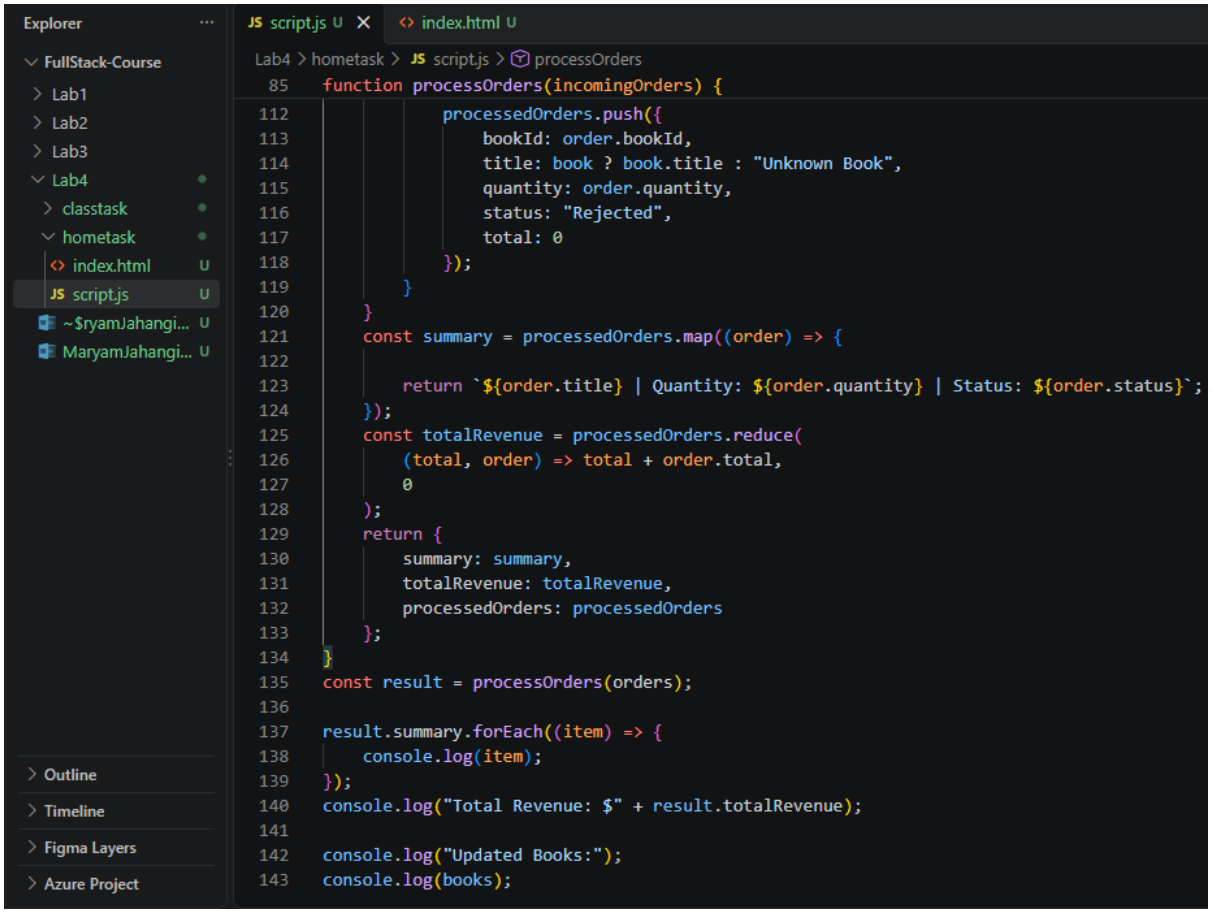
```
3   const books = [
21              author: "James Clear",
22              price: 25,
23              stock: 10,
24              category: "Self Help"  },
25          {
26              id: 4,
27              title: "The Alchemist",
28              author: "Paulo Coelho",
29              price: 20,
30              stock: 4,
31              category: "Fiction"    },
32          {
33              id: 5,
34              title: "Rich Dad Poor Dad",
35              author: "Robert Kiyosaki",
36              price: 30,
37              stock: 6,
38              category: "Finance"   }
39  ];
40  // let because incomingOrders can be changed later
41  let orders = [
42      {
43          bookId: 1,
44          quantity: 2 },
45      {
46          bookId: 3,
47          quantity: 3 },
48      {
49          bookId: 99,
50          quantity: 1 },
51      {
52          bookId: 2,
53          quantity: 5 }
```

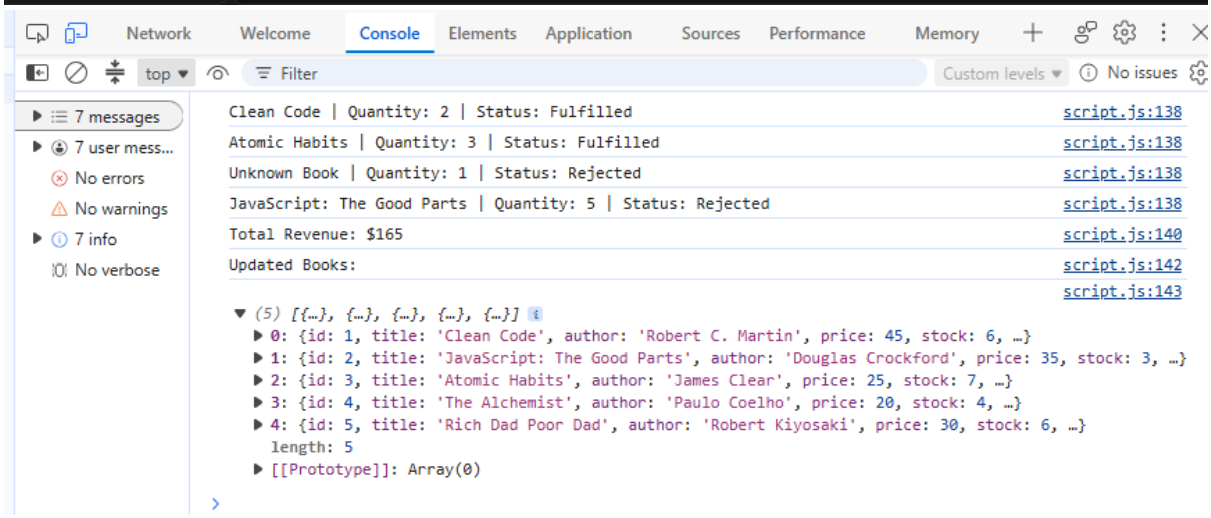

Explorer
script.js U
index.html U
Lab4 > hometask > script.js > ...
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script.js U
~$ryamJahangi... U
MaryamJahangi... U
Outline
Timeline
Figma Layers
Azure Project
47 let orders = [
64];
65 // var example – function scoped and included to demonstrate
66 var bookstoreName = "Online Bookstore";
67 /*
68 console.log("Bookstore:", bookstoreName);
69 console.log("Books:", books);
70 console.log("Orders:", orders);
71
72 console.log(books,5,"Programming"); */
73
74 function validateOrder(order) {
75
76 const book = books.find((b) => b.id === order.bookId);
77
78 const bookExists = book !== undefined;
79 const validQuantity = order.quantity > 0;
80 const enoughStock = bookExists && order.quantity <= book.stock;
81
82 const status =
83 bookExists && validQuantity && enoughStock
84 ? "Order can be fulfilled"
85 : "Order cannot be fulfilled";
86
87 return status;
88 }
89
90 console.log("Valid");
91 console.log(validateOrder(orders[0])); // Valid order
92
93 console.log("Invalid");
94 console.log(validateOrder(orders[2])); // Invalid order


Network
Welcome
Console
Elements
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Performance
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top
Filter
Custom levels
No issues
4 messages
4 user mess...
No errors
No warnings
4 info
No verbose
Valid
script.js:90
Order can be fulfilled
script.js:91
Invalid
script.js:93
Order cannot be fulfilled
script.js:94

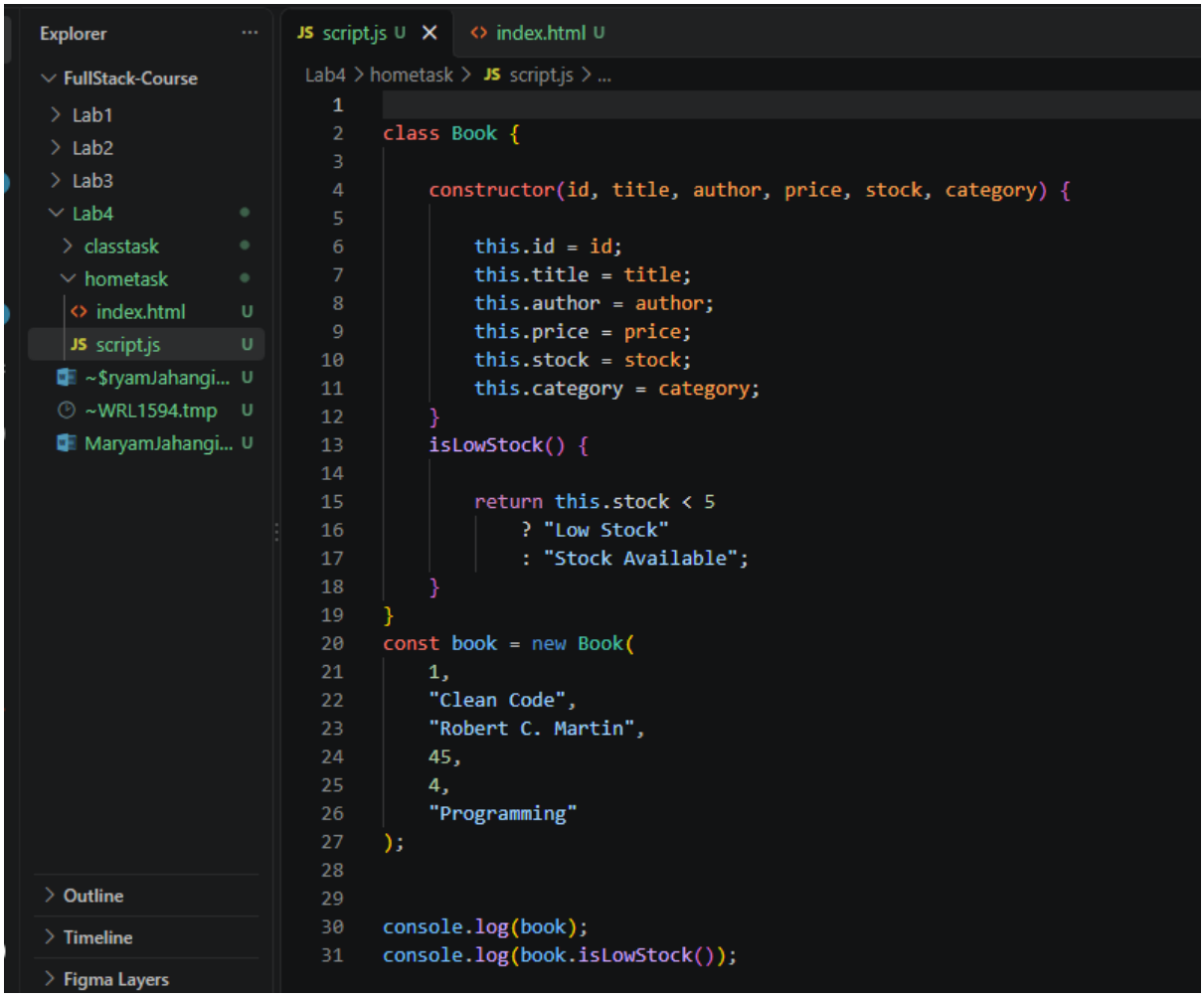
Task 3


Explorer
FullStack-Course
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classtask
hometask
index.html U
script.js U
~$ryamJahangi... U
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script.js U
index.html U
Lab4 > hometask > script.js > processOrders
82
83 console.log("Invalid");
84 console.log(validateOrder(orders[2])); // Invalid order */
85 function processOrders(incomingOrders) {
86
87 const processedOrders = [];
88
89 for (const order of incomingOrders) {
90
91 const book = books.find((b) => b.id === order.bookId);
92
93 const isValid =
94 book !== undefined &&
95 order.quantity > 0 &&
96 order.quantity <= book.stock;
97
98 if (isValid) {
99 book.stock -= order.quantity;
100 const total = book.price * order.quantity;
101
102 processedOrders.push({
103 bookId: book.id,
104 title: book.title,
105 quantity: order.quantity,
106 status: "Fulfilled",
107 total: total
108 });
109
110 } else {
111
112 processedOrders.push({
113 bookId: order.bookId,
114 title: book ? book.title : "Unknown Book",
115 quantity: order.quantity,


Explorer
script.js U
index.html U
Lab4 > hometask > script.js > processOrders
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index.html U
script.js U
~$ryamJahangi... U
MaryamJahangi... U
Outline
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Azure Project
85 function processOrders(incomingOrders) {
112 processedOrders.push({
113 bookId: order.bookId,
114 title: book ? book.title : "Unknown Book",
115 quantity: order.quantity,
116 status: "Rejected",
117 total: 0
118 });
119 }
120 }
121 const summary = processedOrders.map((order) => {
123 return `${order.title} | Quantity: ${order.quantity} | Status: ${order.status}`;
124 });
125 const totalRevenue = processedOrders.reduce(
126 (total, order) => total + order.total,
127 0
128);
129 return {
130 summary: summary,
131 totalRevenue: totalRevenue,
132 processedOrders: processedOrders
133 };
134 }
135 const result = processOrders(orders);
137 result.summary.forEach((item) => {
138 console.log(item);
139 });
140 console.log("Total Revenue: $" + result.totalRevenue);
142 console.log("Updated Books:");
143 console.log(books);

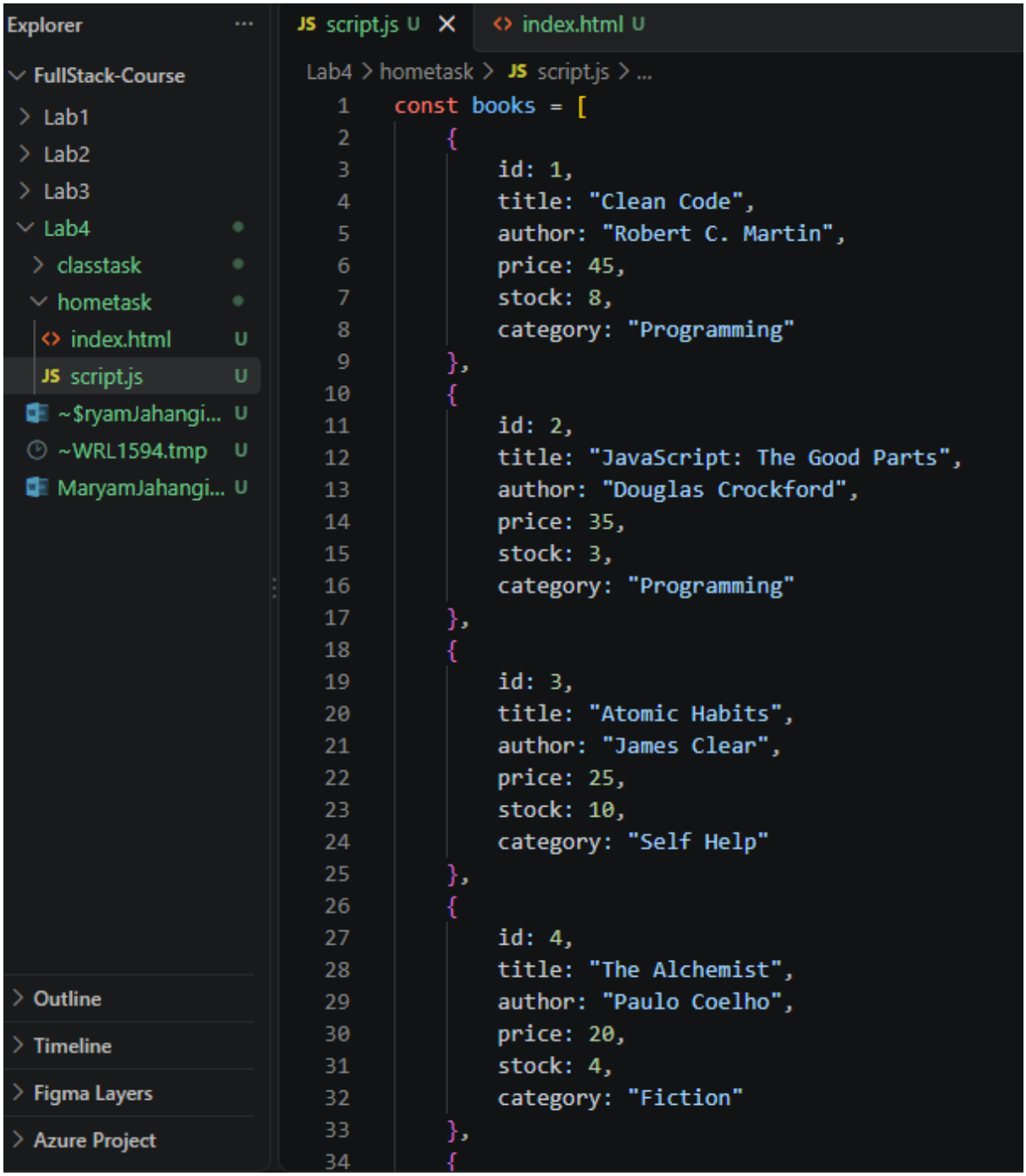

Network
Welcome
Console
Elements
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Sources
Performance
Memory
top
Filter
Custom levels
No issues
7 messages
7 user mess...
No errors
No warnings
7 info
No verbose
Clean Code | Quantity: 2 | Status: Fulfilled script.js:138
Atomic Habits | Quantity: 3 | Status: Fulfilled script.js:138
Unknown Book | Quantity: 1 | Status: Rejected script.js:138
JavaScript: The Good Parts | Quantity: 5 | Status: Rejected script.js:138
Total Revenue: $165 script.js:140
Updated Books: script.js:142
script.js:143
(5) [{…}, {…}, {…}, {…}, {…}]
0: {id: 1, title: 'Clean Code', author: 'Robert C. Martin', price: 45, stock: 6, …}
1: {id: 2, title: 'JavaScript: The Good Parts', author: 'Douglas Crockford', price: 35, stock: 3, …}
2: {id: 3, title: 'Atomic Habits', author: 'James Clear', price: 25, stock: 7, …}
3: {id: 4, title: 'The Alchemist', author: 'Paulo Coelho', price: 20, stock: 4, …}
4: {id: 5, title: 'Rich Dad Poor Dad', author: 'Robert Kiyosaki', price: 30, stock: 6, …}
length: 5
[[Prototype]]: Array(0)

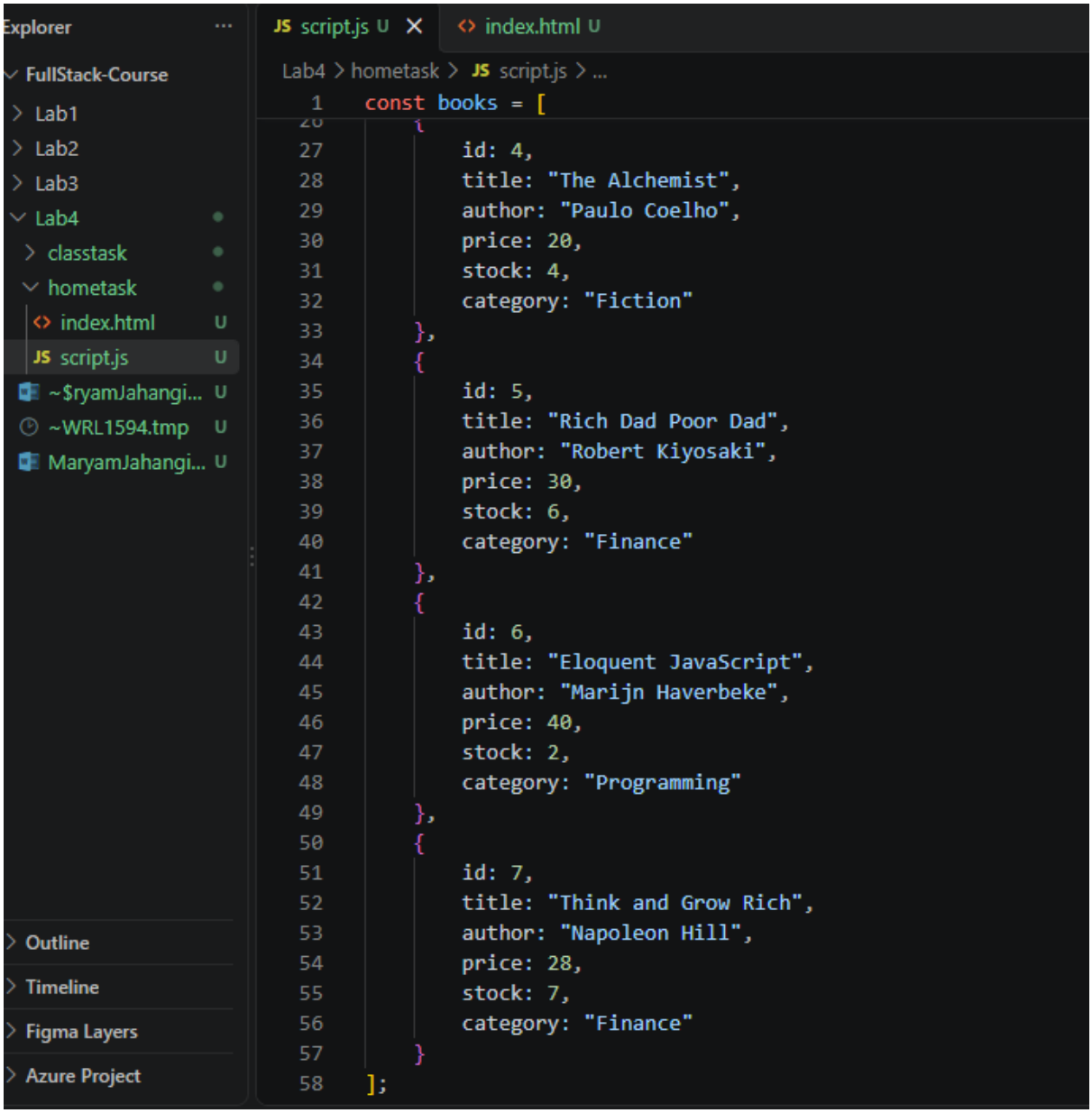
Task 4

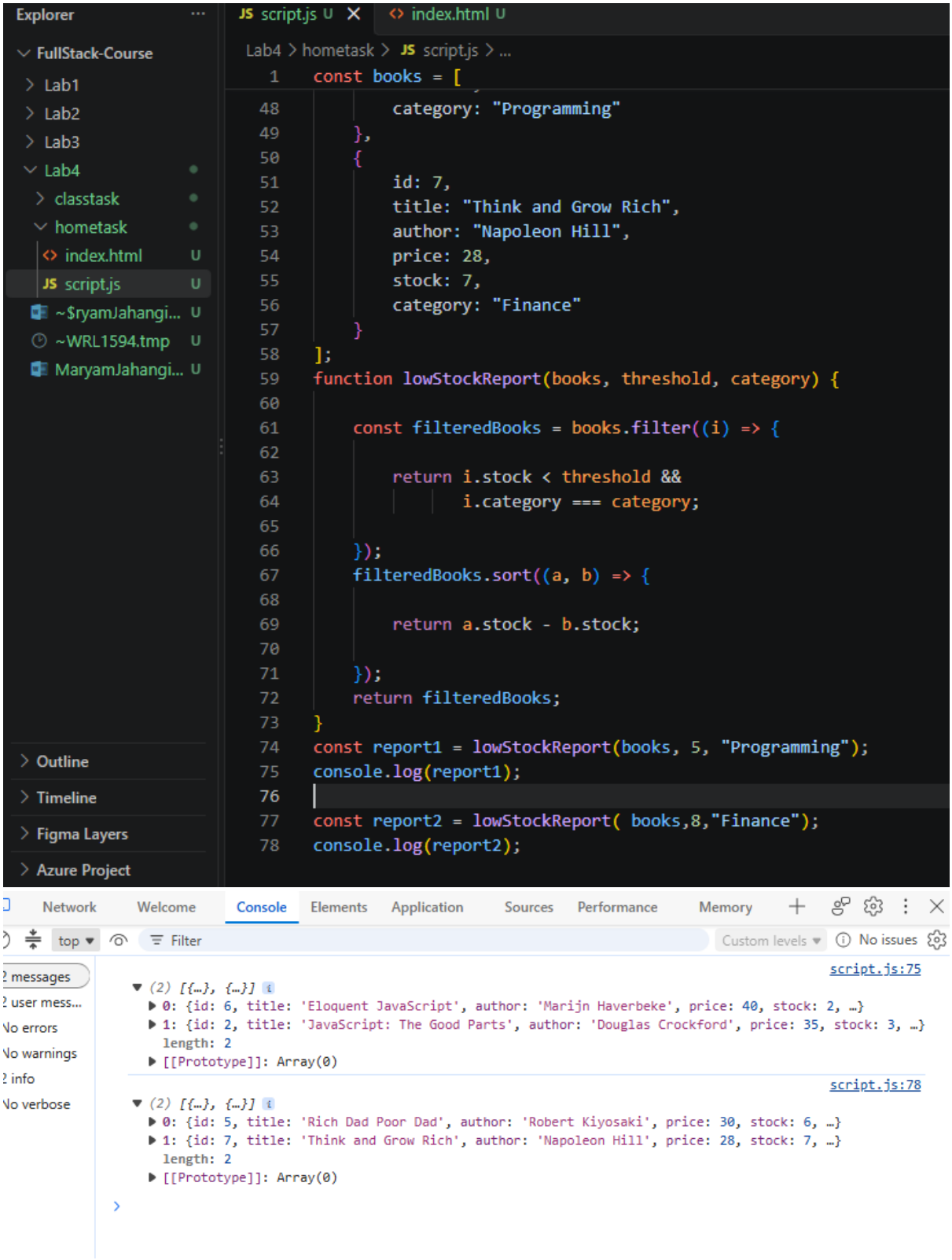

Explorer
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Lab4 > hometask > script.js > ...
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script.js U
~$ryamJahangi... U
~WRL1594.tmp U
MaryamJahangi... U
Outline
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1 class Book {
2
3 constructor(id, title, author, price, stock, category) {
4
5 this.id = id;
6 this.title = title;
7 this.author = author;
8 this.price = price;
9 this.stock = stock;
10 this.category = category;
11 }
12 isLowStock() {
13
14 return this.stock < 5
15 ? "Low Stock"
16 : "Stock Available";
17 }
18 }
19 const book = new Book(
20 1,
21 "Clean Code",
22 "Robert C. Martin",
23 45,
24 4,
25 "Programming"
26);
27
28
29 console.log(book);
30 console.log(book.isLowStock());
31


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Book {id: 1, title: 'Clean Code', author: 'Robert C. Martin', price: 45, stock: 4, …}
script.js:30
Low Stock
script.js:31

Task 5


Explorer
script.js U
index.html U
Lab4 > hometask > script.js > ...
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Lab4
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hometask
index.html U
script.js U
~$ryamJahangi... U
~WRL1594.tmp U
MaryamJahangi... U
Outline
Timeline
Figma Layers
Azure Project
1 const books = [
2 {
3 id: 1,
4 title: "Clean Code",
5 author: "Robert C. Martin",
6 price: 45,
7 stock: 8,
8 category: "Programming"
9 },
10 {
11 id: 2,
12 title: "JavaScript: The Good Parts",
13 author: "Douglas Crockford",
14 price: 35,
15 stock: 3,
16 category: "Programming"
17 },
18 {
19 id: 3,
20 title: "Atomic Habits",
21 author: "James Clear",
22 price: 25,
23 stock: 10,
24 category: "Self Help"
25 },
26 {
27 id: 4,
28 title: "The Alchemist",
29 author: "Paulo Coelho",
30 price: 20,
31 stock: 4,
32 category: "Fiction"
33 },
34 {


Explorer
FullStack-Course
Lab1
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Lab3
Lab4
classtask
hometask
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script.js U
~$ryamJahangi... U
~WRL1594.tmp U
MaryamJahangi... U
Outline
Timeline
Figma Layers
Azure Project
script.js U
index.html U
Lab4 > hometask > script.js > ...
1 const books = [
27 id: 4,
28 title: "The Alchemist",
29 author: "Paulo Coelho",
30 price: 20,
31 stock: 4,
32 category: "Fiction"
33 },
34 {
35 id: 5,
36 title: "Rich Dad Poor Dad",
37 author: "Robert Kiyosaki",
38 price: 30,
39 stock: 6,
40 category: "Finance"
41 },
42 {
43 id: 6,
44 title: "Eloquent JavaScript",
45 author: "Marijn Haverbeke",
46 price: 40,
47 stock: 2,
48 category: "Programming"
49 },
50 {
51 id: 7,
52 title: "Think and Grow Rich",
53 author: "Napoleon Hill",
54 price: 28,
55 stock: 7,
56 category: "Finance"
57 }
58];


Explorer
script.js U
index.html U
Lab4 > hometask > script.js > ...
FullStack-Course
Lab1
Lab2
Lab3
Lab4
classtask
hometask
index.html U
script.js U
~$ryamJahangi... U
~WRL1594.tmp U
MaryamJahangi... U
Outline
Timeline
Figma Layers
Azure Project
1 const books = [
48 category: "Programming"
49 },
50 {
51 id: 7,
52 title: "Think and Grow Rich",
53 author: "Napoleon Hill",
54 price: 28,
55 stock: 7,
56 category: "Finance"
57 }
58];
59 function lowStockReport(books, threshold, category) {
60
61 const filteredBooks = books.filter((i) => {
62
63 return i.stock < threshold &&
64 i.category === category;
65
66 });
67 filteredBooks.sort((a, b) => {
68
69 return a.stock - b.stock;
70
71 });
72 return filteredBooks;
73 }
74 const report1 = lowStockReport(books, 5, "Programming");
75 console.log(report1);
76
77 const report2 = lowStockReport(books,8,"Finance");
78 console.log(report2);
Network
Welcome
Console
Elements
Application
Sources
Performance
Memory
top
Filter
Custom levels
No issues
2 messages
2 user mess...
No errors
No warnings
2 info
No verbose
script.js:75
(2) [{…}, {…}]
0: {id: 6, title: 'Eloquent JavaScript', author: 'Marijn Haverbeke', price: 40, stock: 2, …}
1: {id: 2, title: 'JavaScript: The Good Parts', author: 'Douglas Crockford', price: 35, stock: 3, …}
length: 2
[[Prototype]]: Array(0)
script.js:78
(2) [{…}, {…}]
0: {id: 5, title: 'Rich Dad Poor Dad', author: 'Robert Kiyosaki', price: 30, stock: 6, …}
1: {id: 7, title: 'Think and Grow Rich', author: 'Napoleon Hill', price: 28, stock: 7, …}
length: 2
[[Prototype]]: Array(0)